

Behaviourally-based Synthesis of Scene-aware Footstep Sound

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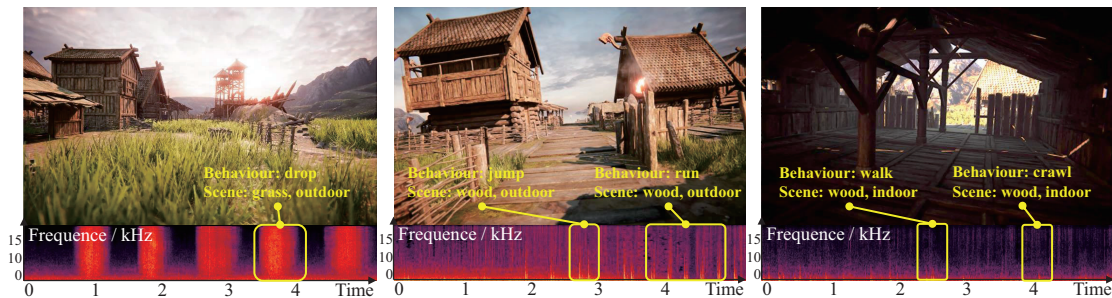


Figure 1: Illustrations of the audiovisual simulation result of roaming in an open scene. The top row presents three frames of the visual simulation results of the user in three different positions. The bottom row shows the sound spectrum of footsteps for the corresponding scene. When the surroundings and behaviour of the user are different, the synthesized footsteps sound vary simultaneously (marked in the yellow box).

ABSTRACT

This paper proposes a behaviourally-based scene-aware (BBSA) footstep sound synthesis method for congruent interaction in virtual environments. Most existing methods focus on how to simulate timbre of footstep sound appropriately, but ignore the impact of user behaviour and virtual scenes on footstep sound, which reduces the sense of immersion and presence. To tackle this issue, we classify common user behaviours and design a set of mapping functions between different behaviours and footsteps. To further enhance user immersion, we propose a ray-casting based scene aware sound synthesis method, which allows to synthesize corresponding footstep sounds in real time for indoor and outdoor scenes with different material surfaces. User studies demonstrate that our proposed BBSA method achieves a higher level of immersion and presence. The complete project code is available: <https://github.com/OlyMarco/Behaviourally-based-Synthesis-of-Scene-aware-Footstep-Sound-Demo->.

Index Terms: Human-centered computing—Human computer interaction—Interaction techniques—Auditory feedback; Applied computing—Arts and humanities—Sound and music computing

1 INTRODUCTION

Real-time footstep synthesis has been a hot research topic in the fields of sound synthesis and virtual reality. In the last two decades, several physical methods [6], physically inspired methods [4, 12] and neural models [3] are proposed for analyzing and parametrically

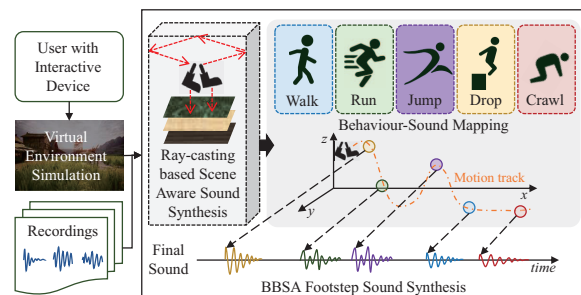


Figure 2: The architecture for BBSA footstep sound synthesis.

synthesizing footstep sound. However, these studies mainly focus on how to generate more timbre-realistic footsteps. To generate interactive footstep sound, researchers explore methods based on the ground reaction force between shoes and the ground [5, 9–11]. Nevertheless, current procedural models of footstep sound simply consider the effect of different ground materials on footsteps [1], which severely decreases the immersion and presence of the user.

In this context, our work strives for a method capable of establishing an efficient mapping between the physical simulation and footstep sound, which can generate different sounds for different behaviours and locations of users. A preliminary use-case scenario is illustrated in Figure 1. Specifically, we propose a behaviourally-based scene-aware (BBSA) footstep sound synthesis method (shown in Figure 2). In each time step, the user inputs the spatial position through the interactive device (mouse, keyboard, etc.). In the sound simulation module, there are two main components in the pipeline: activate and manipulate the corresponding recording based on the detection of the user's current surroundings, modify the sound form the first step according to the user's current behaviour. The final output of the sound simulation module is consistent with the virtual scene and user behaviour.

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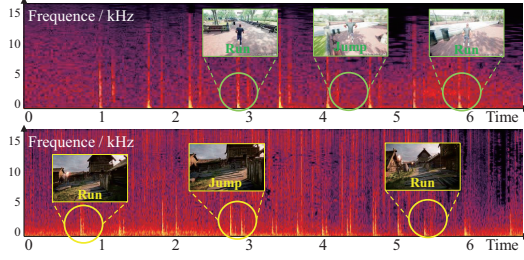


Figure 3: Comparison with the state-of-the-art method. The top row presents the footsteps sound synthesized by the Nemisindo [1], and the bottom row shows the footsteps sound synthesized by our method.

2 BBSA FOOTSTEP SYNTHESIS

2.1 Ray-casting based Scene-aware Sound Synthesis

Towards the goal of congruency within different virtual environments, we design a ray-casting based environment detection method coupled with concatenative sound synthesis [2, 8] to generate scene-aware footstep sound. Specifically, the method consists of two parts, namely material detection and occlusion detection.

Material Detection. Material detection is designed for distinguishing the material of the user's current contact surface and determining the type of recording to be extracted from the sample library. We emit rays from the user's feet to obtain the type of material that the user touches first.

Occlusion Detection. Occlusion detection is mainly designed to achieve differentiation of sounds between indoor and outdoor. Although the Spatial Blend method of Unity allows for indoor reverb, it does not take into account the effect of walls on the sound. Therefore, we traverse all the walls around the user, and form a data chain table consisting of the number of walls and their thickness. Finally, by fitting the decay function, we calculate the volume s_v and pitch s_p values for each sound source.

2.2 Behaviour-Sound Mapping

After synthesizing the original sound according to the user's surroundings, the sound needs to be further modified according to the user's behaviour. For the three behaviours of walking, running and jumping, we divide three states by the distance interval of unit time. For jump and drop, we first detect whether the user is currently on the ground, if not, we calculate whether the user's current height u is larger than the height threshold ω_h .

Once the user behaviour is determined, we modify the footsteps sound by parameters of volume $\{v_r, v_j, v_d, v_c\}$ and pitch $\{p_r, p_j, p_d, p_c\}$ for different behaviours. Taking jump as an example, we first calculate the difference δ between the user's vertical coordinate u_y and height of the user u_h in a short interval. Then δ is utilized as the amplification factor to add $\delta \cdot v_j$ to the volume of the original sound and add $\delta \cdot p_j$ to the pitch of the original sound, where both v_j and p_j can be set by the user. In summary, the mapping of behaviour and sound control is listed as follows.

Walk: $s_v \leftarrow s_v$ and $s_p \leftarrow s_p$;
Run: $s_v \leftarrow s_v + v_r$ and $s_p \leftarrow s_p + p_r$;
Jump: $s_v \leftarrow s_v + \delta \cdot v_j$ and $s_p \leftarrow s_p + \delta \cdot p_j$;
Drop: $s_v \leftarrow s_v + \delta \cdot v_d$ and $s_p \leftarrow s_p + \delta \cdot p_d$;
Crawl: $s_v \leftarrow s_v - v_c$ and $s_p \leftarrow s_p - p_c$.

3 RESULTS AND DISCUSSION

We implement our algorithm on Unity3d game engine and all the experiments are conducted with the same hardware: NVIDIA GeForce GTX 1660 Ti GPU, and 8GB RAM. There are 19 footstep recordings utilized as experimental samples, with a total length of 12 seconds.

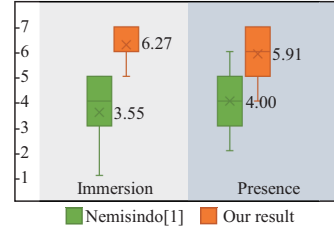


Figure 4: The boxplot shows the user study results comparison of [1] and our results.

In order to validate our BBSA footstep sound synthesis method, we compare it with the state-of-the-art procedural model Nemisindo [1]. As shown in Figure 3, it can be observed that when the user's behaviour is "run-jump-run", Nemisindo cannot differentiate the footsteps of different behaviours, while our method can synthesize different footsteps. Furthermore, we conduct a user study in terms of both immersion and presence, where "presence" is the feeling of "being there" in a virtual environment [7]. Compared to the sound synthesized by [1], our synthesized sound can significantly improve the sense of immersion and presence.

4 CONCLUSIONS AND FUTURE WORK

In this paper, we propose an automatic behaviour mapping for scene-aware footstep sound synthesis. The major contribution of the framework relies in the potential to increase the feeling of presence and immersion in interactive virtual environments.

REFERENCES

- [1] P. Bahadoran, A. Benito, T. Vassallo, and J. Reiss. Fxive: a web platform for procedural sound synthesis. In *Audio Engineering Society 144th Convention*, pp. 1–5, 2018.
- [2] H. Cheng and S. Liu. Haptic force guided sound synthesis in multisensory virtual reality (vr) simulation for rigid-fluid interaction. In *IEEE Virtual Reality*, pp. 111–119, 2019.
- [3] M. Comunità, H. Phan, and J. Reiss. Neural synthesis of footsteps sound effects with generative adversarial networks, 2021. CoRR abs/2110.09605.
- [4] P. R. Cook. Modeling bill's gait: analysis and parametric synthesis of walking sounds. In *International Conference on Virtual, Synthetic and Entertainment Audio*, pp. 1–6, 2002.
- [5] A. J. Farnell. Marching onwards: procedural synthetic footsteps for video games and animation. In *PD Convention*, pp. 1–5, 2007.
- [6] F. Fontana and R. Bresin. Physics-based sound synthesis and control: crushing, walking and running by crumpling sounds. In *XIV Colloquium on Musical Informatics*, pp. 109–114, 2003.
- [7] A. C. Kern and W. Ellermeier. Audio in vr: effects of a soundscape and movement-triggered step sounds on presence. *Frontiers Robotics AI*, 7:1–13, 2020.
- [8] E. Magalhães, J. Jacob, N. C. Nilsson, R. Nordahl, and G. Bernardes. Physics-based concatenative sound synthesis of photogrammetric models for aural and haptic feedback in virtual environments. In *IEEE Virtual Reality*, pp. 376–379, 2020.
- [9] R. Nordahl, S. Serafin, and L. Turchet. Sound synthesis and evaluation of interactive footsteps for virtual reality applications. In *IEEE Virtual Reality*, pp. 147–153, 2010.
- [10] L. Turchet, D. Moffat, A. Tajadura-Jiménez, and J. Reiss. What do your footsteps sound like? an investigation on interactive footstep sounds adjustment. *Applied Acoustics*, 111:77–85, 2016.
- [11] L. Turchet, S. Serafin, S. Dimitrov, and R. Nordahl. Physically based sound synthesis and control of footsteps sounds. In *13th International Conference on Digital Audio Effects*, pp. 161–168, 2010.
- [12] D. L. Yewdall. Foley: the art of footsteps, props, and cloth movement. In *The practical art of motion picture sound*, pp. 402–439. Routledge, 2012.